

Introduction to Electronics Engineering

MODULE 5 — PART 2: QUESTION BANK

Module 5 (08 hrs): Introduction to Electronic Communication Systems

For First-Year Students — with complete answers and diagrams
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 – Q33
Section B	7 marks	Compulsory — from every module	Q1 – Q9

- Module 5 has **no 5-mark section** — 5-mark questions are asked only from Modules 1 and 2.
- Every topic of Module 5 has a 2-mark question, and every topic is also covered in the 7-mark section — no topic is left out.
- No question is repeated between sections.
- Answer 2-mark questions in 2–4 lines and 7-mark questions with full explanation and diagrams.

SECTION A — 2-Mark Questions

Q1. What is the electromagnetic spectrum? Name its bands in order.

Answer: The electromagnetic spectrum is the complete range of electromagnetic waves arranged by their frequency and wavelength. From longest wavelength to shortest: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays.

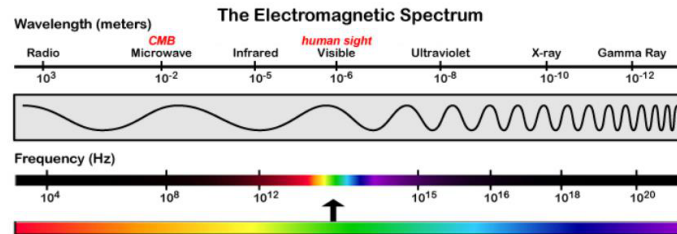


Figure 1: The electromagnetic spectrum.

Q2. Give the applications of any four bands of the electromagnetic spectrum.

Answer: Radio waves — broadcasting and mobile communication; microwaves — radar, satellite links and Wi-Fi; infrared — remote controls and optical fibre; X-rays — medical imaging and security scanning.

Q3. Write the relation between wavelength and frequency of an electromagnetic wave.

Answer: $\lambda = c / f$, where λ is the wavelength, $c = 3 \times 10^8$ m/s is the speed of light, and f is the frequency. Low frequency means long wavelength; high frequency means short wavelength.

Q4. Write the elements of a basic communication system.

Answer: Input transducer, transmitter, transmission channel, receiver and output transducer.

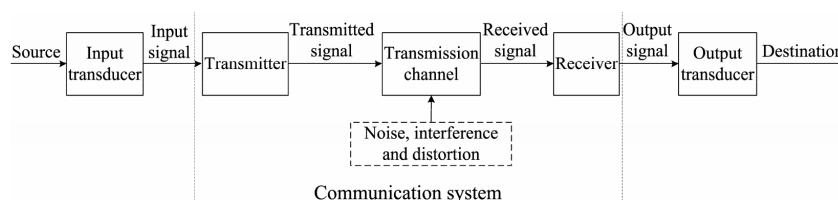


Figure 2: Block diagram of a communication system.

Q5. What is a transducer? Give two examples.

Answer: A transducer is a device that converts energy from one form to another, typically converting a physical quantity into an electrical signal or vice versa. Examples: microphone (sound to electricity) and speaker (electricity to sound).

Q6. What is the difference between an input transducer and an output transducer?

Answer: An input transducer (sensor) converts a physical parameter into an electrical signal; an output transducer (actuator) converts an electrical signal into a physical action.

Q7. What is an analog message? Give two examples.

Answer: An analog message is continuous in amplitude and over time. Examples: AM/FM voice or sound, traditional TV video, and the output of a record player.

Q8. What is a digital message? Give two examples.

Answer: A digital message takes discrete values (0 or 1). Examples: data on a disk and VCD/DVD, and signals in a cellular mobile phone.

Q9. What operations are performed by a transmitter?

Answer: The transmitter encodes the message data and adds a carrier signal (modulation), so that the message becomes suitable for transmission over the channel.

Q10. What operations are performed by a receiver?

Answer: The receiver receives the signal, removes the carrier (demodulation), decodes the data and removes noise, in order to recover the original message.

Q11. What is a communication channel? Give two examples.

Answer: The communication channel is the medium by which an electronic signal is transmitted from one place to another. Examples: conducting wires, coaxial cable, optical cable and free space.

Q12. What is the difference between wireline and wireless communication?

Answer: In wireline communication (e.g. telephone, fibre-optic) two physical wires connect the transmitter and receiver; in wireless communication (e.g. mobile, radio, satellite) no physical wires are used — the signal travels through free space.

Q13. Define modulation.

Answer: Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal, with a modulating signal that contains the information to be transmitted.

Q14. What are the message signal and the carrier signal? Which is low and which is high frequency?

Answer: The message signal $m(t)$ (also called the baseband or modulating signal) is a low-frequency signal that contains the information. The carrier signal $c(t)$ is a high-frequency sinusoidal or square-pulse signal used to carry it.

Q15. What is amplitude modulation (AM)?

Answer: AM is the process of varying the amplitude of a carrier wave in proportion to the amplitude of the baseband signal, while the frequency of the carrier remains constant.

Q16. What is frequency modulation (FM)?

Answer: FM is the process of varying the frequency of a carrier wave in proportion to the amplitude of the baseband signal, while the amplitude of the carrier remains constant.

Q17. What is phase modulation (PM)?

Answer: PM is a form of analog modulation in which the phase of the carrier wave is varied in proportion to the baseband signal.

Q18. Why is modulation needed? Give the four reasons.

Answer: 1. Reduction in the height of the antenna; 2. Increases the range of communication; 3. Improves the quality of reception; 4. Multiplexing is possible.

Q19. Why is antenna height important? Write the relation.

Answer: For radio transmission the antenna height must be at least $\lambda/4$, where $\lambda = c/f$. For a low-frequency baseband signal this height is impractically large (about 7.5 km for 10 kHz), which is why we modulate to a higher frequency.

Q20. What is multiplexing?

Answer: Multiplexing is the process in which two or more signals can be transmitted over the same communication channel simultaneously. This is possible only with modulation, because each signal is placed on a different carrier frequency.

Q21. What is IoT?

Answer: The Internet of Things (IoT) is a system of interrelated, internet-connected objects that can collect and transfer data over a wireless network without human intervention.

Q22. Give the different names of IoT.

Answer: Internet of Everything, Smarter Planet, Machine to Machine (M2M), Tsensors (Trillion Sensors), the Industrial Internet, Industry 4.0 and Internet of Things.

Q23. What are the four components (stages) of IoT?

Answer: Sensors/Devices, Connectivity, Data Processing and User Interface.

Q24. What is the role of sensors in IoT?

Answer: Sensors connect with the external physical environment; they sense changes in the surroundings, collect the data and store it. For example, the GPS sensor in a phone tracks location and guides you to your destination.

Q25. What is connectivity in IoT?

Answer: Connectivity is the stage in which the collected data is sent to a cloud infrastructure using various communication mediums such as mobile or satellite networks, Bluetooth, Wi-Fi and WAN.

Q26. What is data processing in IoT?

Answer: Once the data reaches the cloud, software processes it — from simple checks (temperature, AC/heater readings) to complex tasks such as object identification using computer vision on video.

Q27. What is the user interface in IoT?

Answer: The user interface makes the information available to the end-user — e.g. by triggering alarms on phones or sending notifications through email or text message, or through a web server.

Q28. State any three advantages of IoT.

Answer: It eliminates/reduces human error; decreases the wastage of resources (time, material, effort); and increases security and accuracy (also boosts productivity and enhances quality).

Q29. State any three disadvantages of IoT.

Answer: If the interconnected devices are hacked the entire system security is jeopardised; the initial hardware investment is heavy; and a fault in one device may disrupt the whole system.

Q30. What does 'G' stand for in 1G to 5G?

Answer: 'G' stands for **Generation** — a generation of wireless phone technology.

Q31. Write the data rate of each mobile generation (1G to 5G).

Answer: 1G: 2 kbps; 2G: 64 kbps; 3G: 144 kbps–2 Mbps; 4G: 2–100 Mbps; 5G: more than 1 Gbps.

Q32. What is the frequency band of 5G?

Answer: 5G uses 450 MHz–6 GHz and 24.25–52.6 GHz (millimetre-wave bands).

Q33. Name any four applications of IoT.

Answer: Smart homes, smart wearables (fitness bands/watches), driverless cars and drones, and smart healthcare monitoring.

SECTION B — 7-Mark Questions

Q1. Explain the electromagnetic spectrum in detail. Name its bands, explain the frequency-wavelength relation, and give the applications of each band.

Answer:

Electromagnetic waves are made of an oscillating electric field and an oscillating magnetic field travelling together through space. They need no material medium, and all travel at the speed of light $c = 3 \times 10^8$ m/s in vacuum.

The **electromagnetic spectrum** is the complete family of these waves arranged by frequency and wavelength. Wavelength and frequency are linked by $\lambda = c/f$: a low frequency gives a long wavelength, and a high frequency gives a short wavelength. In order from longest wavelength to shortest, the bands are radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays.

Band	Application
Radio waves	Broadcasting (AM/FM), television, mobile communication
Microwaves	Radar, satellite links, Wi-Fi, microwave ovens
Infrared	Remote controls, thermal imaging, optical fibre
Visible light	Vision, optical fibre, photography

Band	Application
Ultraviolet	Sterilisation, detecting forged notes
X-rays	Medical imaging, security scanning
Gamma rays	Cancer treatment, sterilising equipment

Thus each band of the spectrum has its own characteristics and its own set of practical uses.

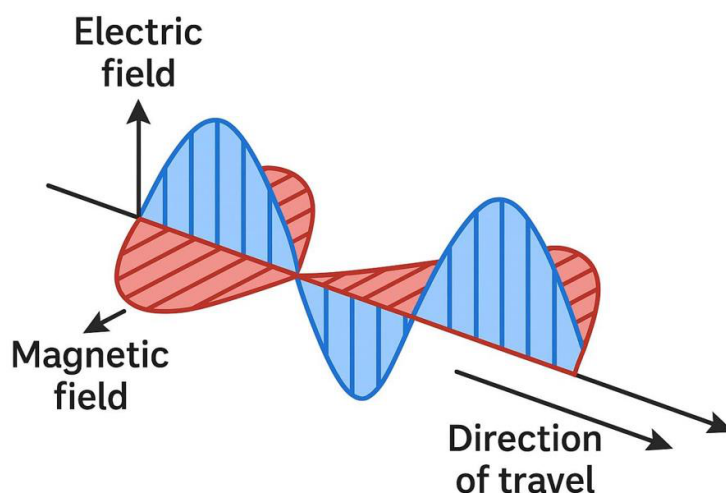


Figure 3: An electromagnetic wave.

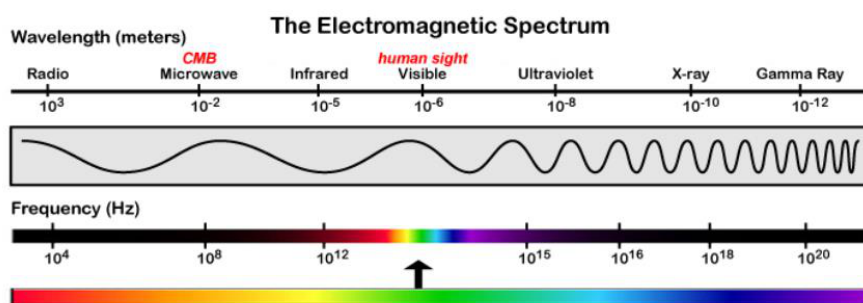


Figure 4: The electromagnetic spectrum and its applications.

Q2. Explain the elements of an electronic communication system with a block diagram, and explain each element.

Answer:

A communication system carries information from a source to a destination through five stages.

- **Input transducer:** converts the message (sound, picture, data) into an electrical signal.
- **Transmitter:** converts the input signal into a transmitted signal suited to the channel — it encodes the data and adds the carrier (modulation).
- **Transmission channel:** the electric medium that bridges the distance from source to destination.
- **Receiver:** converts the received signal into a form appropriate for the output transducer — it demodulates, decodes and removes noise.
- **Output transducer:** converts the output electrical signal back into the desired message form.

The order is therefore: message → transducer → transmitter → channel → receiver → transducer → message. The channel may be **wireline** (telephone, fibre-optic — physical wires connect the ends) or **wireless** (mobile, radio, satellite — no physical wires).

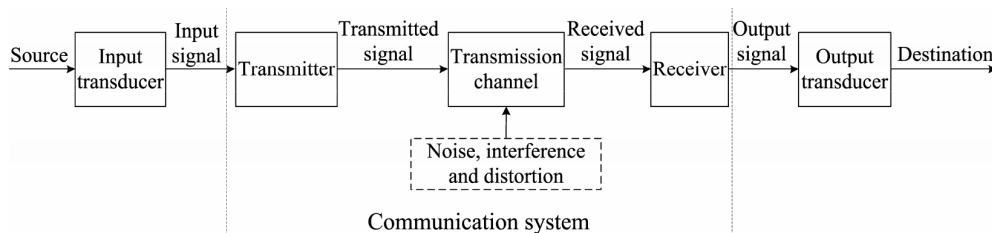


Figure 5: Block diagram of a communication system.

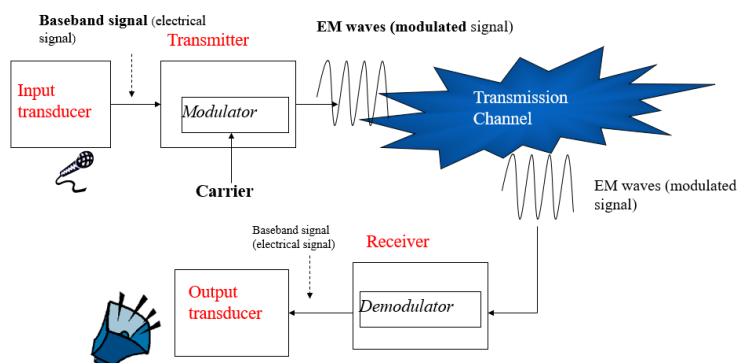


Figure 6: Transmitter and receiver arrangement.

Q3. Explain the components of a communication system — the message (analog and digital) and the transducer.

Answer:

The message comes from the message source and may be analog or digital:

Analog Message	Digital Message
Continuous in amplitude and over time	Discrete value (0 or 1)
AM, FM for voice/sound	Cellular mobile phone
Traditional TV for analog video	Data on your disk
Record player	VCD, DVD

An analog message can take any value in a range, so noise disturbs it easily. A digital message has only two levels, so it can still be recognised correctly even if noise slightly changes the voltage — this is why modern systems are digital.

The transducer is a device that converts energy from one form to another. Input transducers (sensors) convert physical parameters into electrical signals; output transducers (actuators) convert electrical signals into physical action. Common examples: microphones, speakers, thermometers, solar cells and ultrasonic probes.

Q4. Explain modulation. What are the message and carrier signals? Explain AM, FM and PM with waveforms.

Answer:

Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal, with a modulating signal that contains the information. We load the message onto a high-frequency carrier so that it can travel through the channel — just as a light parcel is carried by a powerful truck.

- $m(t)$ — the **message/baseband signal**, a low-frequency signal (audio 300 Hz–3.4 kHz, video 0–5 MHz).

- $c(t)$ — the **carrier signal**, a high-frequency sinusoid or square pulse.
- $s(t)$ — the **modulated signal** that is actually transmitted.

A sinusoidal carrier has amplitude, frequency and phase; varying each one gives the three analog modulation techniques:

- **AM:** the amplitude of the carrier is varied in proportion to the baseband signal; the frequency stays constant.
- **FM:** the frequency of the carrier is varied in proportion to the baseband signal; the amplitude stays constant.
- **PM:** the phase of the carrier is varied in proportion to the baseband signal.

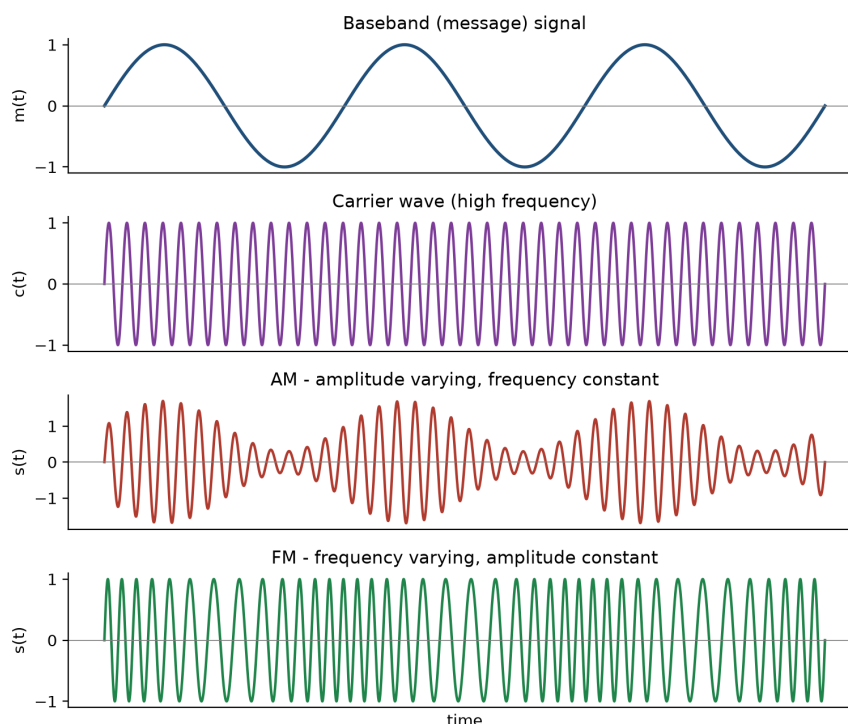


Figure 7: Baseband, carrier, AM and FM waveforms.

Q5. Explain the need for modulation in detail, with the antenna-height calculation.

Answer:

Modulation is necessary for four reasons, and each is explained below:

(a) Reduction in the height of the antenna: the antenna height must be at least $\lambda/4$, where $\lambda = c/f$. To transmit a baseband signal of 10 kHz directly, $\lambda = 3 \times 10^8 / 10^4 = 30,000$ m, so $\lambda/4 = 7,500$ m — an antenna of this height is impossible to install. Modulating onto a higher-frequency carrier shortens λ and makes the antenna practical.

(b) Increases the range of communication: low-frequency baseband signals are heavily attenuated and cannot travel long distances; modulation raises the transmitted frequency, which travels further.

(c) Improves the quality of reception: FM and digital techniques such as PCM greatly reduce the effect of noise.

(d) Multiplexing is possible: two or more signals can be transmitted over the same channel at the same time when they are modulated onto different carriers.

$$\text{Minimum antenna height} = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3 \times 10^8}{4 \times 10 \times 10^3} = 7500 \text{ meters i.e. } 7.5 \text{ km}$$

Figure 8: Antenna-height calculation.

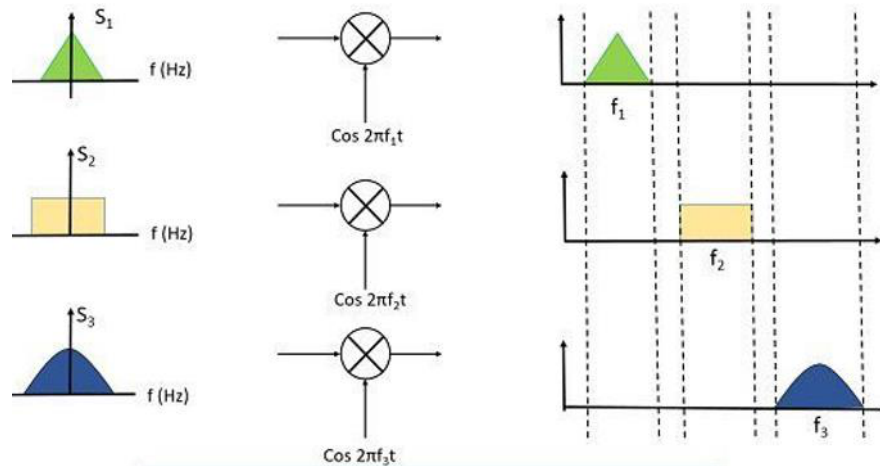


Figure 9: Frequency translation of several signals (basis of multiplexing).

Q6. What is IoT? Give its different names and explain how IoT works.

Answer:

The **Internet of Things (IoT)** is a system of interrelated, internet-connected objects that can collect and transfer data over a wireless network **without human intervention**. It consists of a large network of interconnected devices — smart wearables, driverless cars, drones, smart homes and smart cars.

Different names of IoT: Internet of Everything, Smarter Planet, Machine to Machine (M2M), Tsensors (Trillion Sensors), the Industrial Internet, Industry 4.0 and Internet of Things.

How IoT works — four stages:

- 1 Sensors/Devices sense the environment and collect data.
- 2 Connectivity sends the data to the cloud.
- 3 Data Processing analyses the data and makes it useful.
- 4 User Interface delivers the useful information to the user.

Sensors feed data to IoT platforms, which gather the information and detect patterns; this helps organisations reduce cost through improved process efficiency, asset utilisation and productivity.

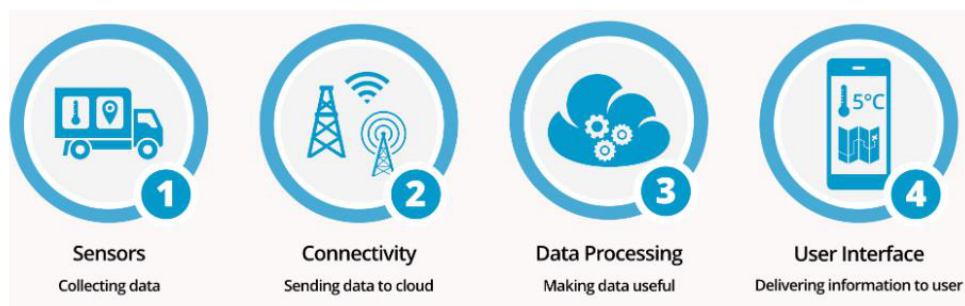


Figure 10: The four stages of IoT.

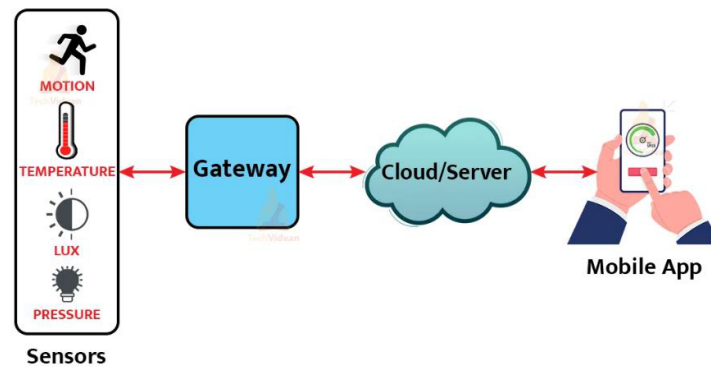


Figure 11: IoT data flow: sensors → gateway → cloud → app.

Q7. Explain the components of IoT in detail.

Answer:

- 1. Sensors:** these devices connect with the external physical environment, sense the changes in the surroundings, collect the data and store it. This property makes sensors extremely useful in IoT. For example, the GPS sensor in a phone tracks location and guides you to your destination.
- 2. Connectivity:** all the collected data is sent to a cloud infrastructure using various communication mediums such as mobile or satellite networks, Bluetooth, Wi-Fi and WAN.
- 3. Data Processing:** once the data reaches the cloud, software processes it — from simple checks (temperature, AC/heater readings) to complex tasks such as object identification using computer vision on video.
- 4. User Interface:** the information is made available to the end-user — by triggering alarms on phones, sending notifications by email or text, or through a web-server interface (as in a home camera system).

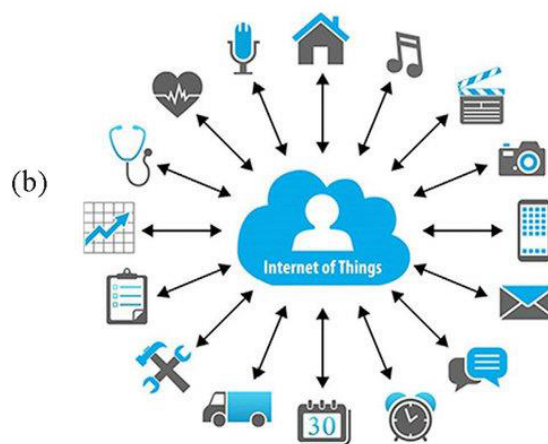
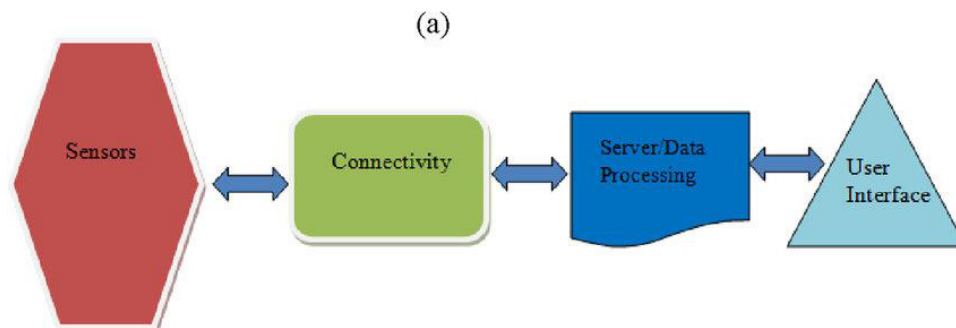


Figure 12: Structure of the Internet of Things and types of sensors.

Q8. Explain the advantages, disadvantages and applications of IoT.**Answer:****Advantages:**

- Eliminates or reduces the chances of human error.
- Decreases the wastage of resources — time, material and human effort.
- Boosts confidence and productivity.
- Increases security and accuracy, and enhances quality.

Disadvantages:

- If the interconnected devices are hacked, the entire system security may be jeopardised.
- The initial investment on hardware is heavy.
- It will absorb many jobs that were done manually.
- A fault in one device may disrupt the entire system.
- Too much dependence on IoT decreases confidence in human effort.

Applications: smart homes, smart wearables, healthcare monitoring, driverless cars and drones, industrial automation, agriculture and smart cities.

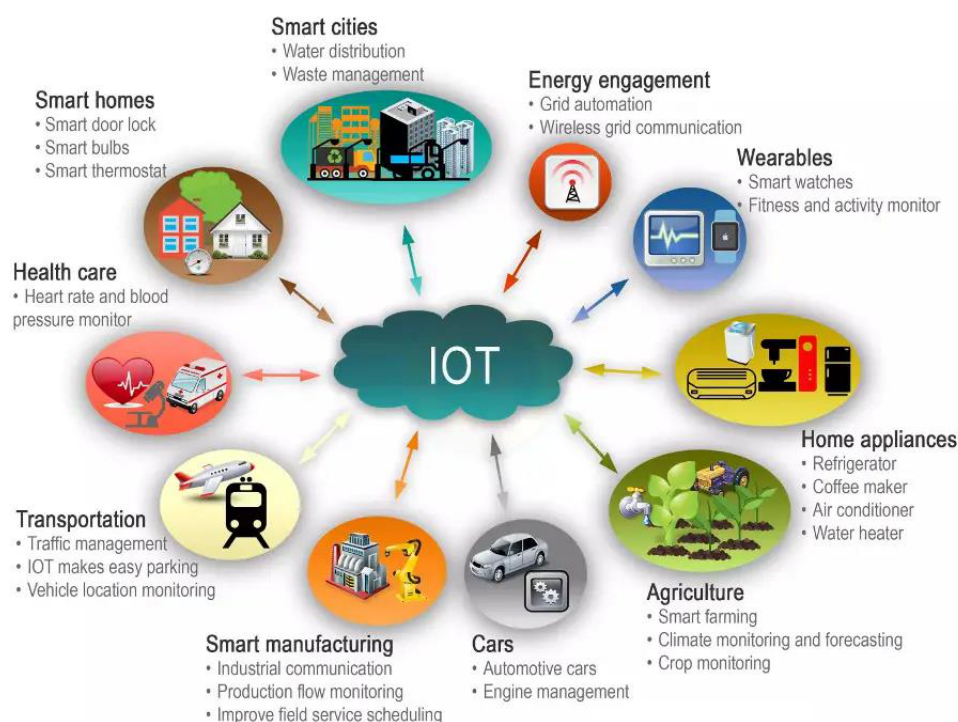


Figure 13: Common application areas of IoT.

Q9. Explain the evolution of mobile communication from 1G to 5G with a comparison table.**Answer:**

1G (1980–1990): the first wireless communication; analog cellular; 2 kbps; bad voice quality, poor battery, big cellphones.

2G (1991–2000): digital cellular (GSM); 64 kbps; text messaging; 2.5G added GPRS for e-mail, web browsing and camera phones.

3G (2000–2010): digital broadband with increased speed (CDMA, UMTS, EDGE); 144 kbps–2 Mbps; smartphones, video calls, mobile TV.

4G (2010–today): high speed, all-IP (LTE, Wi-Fi); 100 Mbps–1 Gbps; MAGIC features (Mobile multimedia, Anytime anywhere, Global mobile support, Integrated wireless, Customized service).

5G (2020 onwards): data rate more than 1 Gbps; high speed and capacity; supports interactive multimedia, voice streaming and IoT-based communication.

Feature	1G	2G	3G	4G	5G
Data rate	2 kbps	64 kbps	144 kbps–2 Mbps	2–100 Mbps	> 1 Gbps
Multiplexing	FDMA	TDMA/CDMA	WCDMA	OFDM	OFDM, NOMA, FBMC
Switching	Circuit	Circuit	Circuit/Packet	Packet	Packet
Application	Voice	Voice, SMS	Voice, data, video	High-quality video	IoT communication

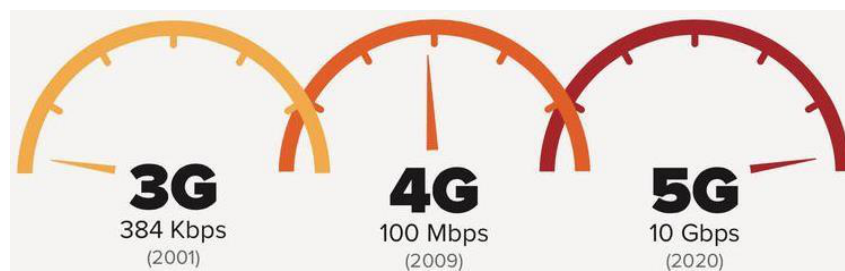


Figure 14: Data-rate growth across generations.

Evolution of mobile phone communications

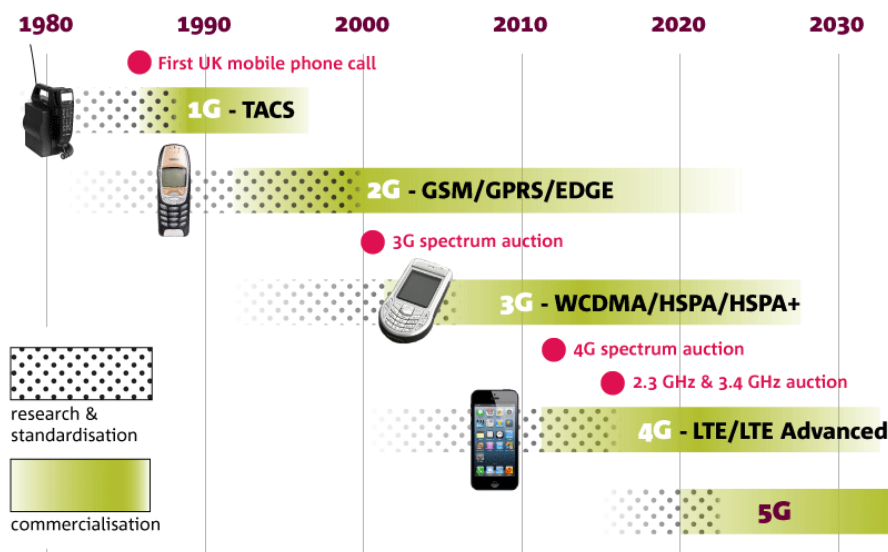


Figure 15: Timeline of mobile communication evolution.